

CO₂ Capture & Mineralisation Plant

1 tonne CO₂/day · skid-mounted · MEA capture + mineralisation to CaCO₃ & K₂SO₄

Estimate class AACE 4 (±30%) **Basis** £2024, equipment + installation **Prepared** Fractional Forge · ForgeOS

Now mineralises the full 1 t/day

INSTALLED PLANT CAPITAL (CENTRAL)

£3.2 M

range £1.8 M – £4.9 M · with the second carbonation sink

CAPTURE COST

£8,700/t-yr

mid-band vs IEA DAC

ALL-IN (FUNDABLE)

~£4.1 M

+ contingency + EPC

What changed in this update. Two things. **(1) Design:** a **second carbonation sink** has been added so the plant mineralises the **full 1 t/day** of captured CO₂ — the primary gypsum route is calcium-limited and fixes only ~0.8 t/day on its own. **(2) Numbers:** every equipment line is now costed from published cost curves (DOE/NETL-2002/1169) on a traceable basis, replacing the earlier model-authored figures. The headline installed cost moves from ~£2.6 M to **~£3.2 M central** — the ~£0.5 M rise is the second sink.

1 · Design update — the second carbonation sink

The calcium balance sets the issue. The MEA system captures the full **1 t/day CO₂** (≈22.7 kmol/day) and strips it for mineralisation. But the primary gypsum route supplies only the calcium in **3.1 t/day of gypsum** (≈18 kmol/day Ca), which can fix **~0.8 t/day CO₂** as ~1.8 t/day CaCO₃. The brief's target of 2.3 t/day CaCO₃ needs ~23 kmol/day Ca — so ~0.2 t/day of captured CO₂ has no calcium to react with.

The **second carbonation sink** closes that gap: a separate carbonation stage with its own alkaline-cation feed (supplementary lime / Ca–Mg source) mineralising the balance **~0.2 t/day CO₂ → ~0.5 t/day additional CaCO₃**. It is sized at ~28% of the primary train — a contained addition (a second reactor + cation feed handling + solids separation/drying), **not** a doubling of the plant. With it, the plant mineralises 100% of captured CO₂ rather than venting the ~20% balance after stripping.

One number to confirm. The brief is internally inconsistent — 3.1 t/day gypsum yields ~1.8 t/day CaCO₃ stoichiometrically, not the stated 2.3 t/day. The second sink is sized to that ~0.5 t/day CaCO₃ gap. If instead the intent is to raise the **primary** gypsum feed (more gypsum → more CaCO₃ + K₂SO₄), the cost lands similarly but the feedstock and product tonnages rise. Confirm the cation source for the second sink before quotes — it sets ~£0.2 M of the estimate.

2 · Purchased-equipment cost (PEC) by section

SECTION	PEC (£)	NOTES
Skid & structure	56,000	2 modules + plinth, ~16 t steel
Thermal utilities	100,800	steam boiler, cooling skid, instrument air (small-package, below curve floors)
MEA absorption	120,400	absorber + packing + blower + quench
MEA recovery	138,500	stripper + reboiler + condenser + coolers + pumps
Carbonation reactor (primary)	112,400	1.0 m³ 316L + agitator + gypsum/KOH feed
CaCO ₃ filtration + drying	71,200	centrifuge + rotary dryer
K ₂ SO ₄ crystallisation	207,000	FC crystalliser (RFQ) + filter + dryer
Bagging / packaging	81,000	25 kg bagging line
Second carbonation sink NEW	170,000	2nd reactor (~0.4 m³) + cation feed handling + solids separation/drying for ~0.5 t/d CaCO ₃
Electrical / instruments / control / safety	in factor	folded into the installation factor (\$4) — not double-counted
Purchased equipment total	1,057,000	range £0.73 M – £1.40 M

3 · Cost-basis trail — major lines (where each number comes from)

EQUIPMENT	£	BASIS & WORKING (TRACEABLE)	FLAG
Absorber column 316L, DN300 × 16.5 m	61,200	Packed Column 15 psig (DOE/NETL p12) · 12 m packed = 39 ft on ID-1 ft line → \$13k (1998\$, extrapolated) × 2.054 CEPCI × 2.90 (316SS) × 0.79	RFQ
Stripper column 316L, DN300 × 12 m	51,800	Packed Column 15 psig (p12) · 8 m packed = 26 ft, ID-1 ft → \$11k × 2.054 × 2.90 × 0.79	RFQ
Carbonation reactor (primary) 316L, 1.0 m³ + 4 kW agitator	82,200	Vertical Vessel p5 + Agitator p27 · 264 gal \$8.5k + 5.4 hp \$9k → × 2.054 × 2.90 × 0.79	CURVE
Carbonation reactor (2nd sink) 316L, ~0.4 m³ + 2 kW agitator	~50,000	Vertical Vessel p5 + Agitator p27 · 106 gal + 2.7 hp, same chain. Near the curve floor at this size → quote.	NEW · RFQ
K₂SO₄ crystalliser forced-circulation package	50–150k	No curve fits a packaged crystalliser. RFQ band; the single dominant uncertainty.	RFQ
Lean/rich exchanger plate, 8 m²	15,400	Spiral-Plate HX p17 (SS304) · 86 ft² \$9.5k × 2.054 × 0.79 (no alloy factor)	CURVE
Reboiler / condenser ~3 m² each	~15–25k ea	Shell-&-Tube HX p15 — below the 100 ft² curve floor; small-kettle/plate figure, not the inflated chart minimum → quote	RFQ
MEA circulation pump 0.68 m³/h	4,000	Live catalogue price — Grundfos CRNE 5-series (pump curve bottoms out above this duty → catalogue is the right basis)	QUOTE

EQUIPMENT	£	BASIS & WORKING (TRACEABLE)	FLAG
Flue-gas blower 225 m³/h, 1.5 kW	21,100	Rotary Blower 8 psig p36 · 132 cfm \$13k × 2.054 × 0.79 (carbon-steel gas service)	CURVE

Every line carries its method, inputs, factors, source page, estimate class and an RFQ flag — the full structured record is generated automatically per line by the cost engine (process-equipment-cost.ts, 20/20 tests).

4 · From equipment to installed cost

Installed cost = purchased equipment × an installation factor. For this **shop-fabricated skid plant** the factor is **2.5 (low) / 3.0 (central) / 3.5 (high)** — well below the stick-built fluids-plant figure of 4.74, because shop labour replaces field labour, piping and wiring are pre-installed on the modules, and the site is a plinth not a foundation field. The two **field-erected 316L columns** plus PED/ATEX duty push toward the top of that band. Electrical, instruments, control and safety are folded into the factor (not added separately, so no double-count).

CASE	INSTALLED CAPITAL	£/(T·YR CO ₂)
Low (PEC £0.73 M × 2.5)	£1.82 M	£4,990
Central (PEC £1.06 M × 3.0)	£3.17 M	£8,690
High (PEC £1.40 M × 3.5)	£4.89 M	£13,400

£3.2 M is the installed plant, not the fundable figure. It excludes contingency (Class-4: +15–30%), engineering/EPC fee (+10–15%), owner's cost, land/buildings, first MEA fill and spares. **All-in plant capital ≈ £4.1 M** before land and owner's cost. If a £2.6 M number is in front of investors as "build cost", it is missing the contingency they will expect — close that gap before it is scrutinised.

5 · What firms the number — quote these

These take the estimate from ±30% to ~±13%, and on balance bias the number **upward**:

1. **K₂SO₄ forced-circulation crystalliser** (£100k ±50%) — no curve fits a packaged unit; the single dominant uncertainty. Quote first.
2. **The two field-erected 316L columns** (£133k, extrapolated curve) — real tall columns cost more than naïve reads; the fabricator quote also settles the solid-316 vs clad-lined question (the biggest equipment-cost lever).
3. **The skid-build / installation factor** (3.0 vs 3.5 = ±£0.5 M) — an EPC/skid-builder quote on the factor itself is the largest single lever on the total.
4. **The second carbonation sink** (~£170k) — its scope depends on the confirmed cation source (lime vs more gypsum); quote once that's set.
5. Solids trains and bagging — could come **down** if shared between the two product streams.

6 · Confidence & sensitivities

Confidence: moderate on the £3.2 M aggregate, low on any single line. At 1 t/day most equipment sits below the public cost-curve floors, so individual lines are rough — but the errors are not one-directional, so the total is robust. The £8,700/(t·yr) sits mid-band against the IEA DAC reference (£1,500–10,000/(t·yr)), independently corroborating the figure.

Dominant sensitivities, in order: (1) the installation factor (2.5↔3.5, ±£0.6 M); (2) the crystalliser package (no curve); (3) solid-316 vs clad-lined metallurgy on the columns and reactors; (4) the second-sink cation source; (5) the below-floor small equipment (boiler, cooling, small HX, dryers) collectively ~£250k of curve-floor substitutions.

Sources. Equipment cost curves: H.P. Loh, J. Lyons, C.W. White III, *Process Equipment Cost Estimation, Final Report*, DOE/NETL-2002/1169 (1Q-1998 US\$). Escalation: CEPCI 1998 = 389.5 → 2024 ≈ 800. Alloy factors: DOE/NETL Table 7. USD→GBP 0.79. Installation: skid-modular factor 2.5–3.5. Sizing: independent first-principles concept sizing (Towler & Sinnott / Perry's). Industry band: IEA DAC 2024.

AACE Class 4 (±30%) concept estimate — defensible and traceable, but estimates not vendor quotes. Quoting the items in §5 takes it to ±13%. Prepared by Fractional Forge · ForgeOS.